LAB 2

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```
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
```

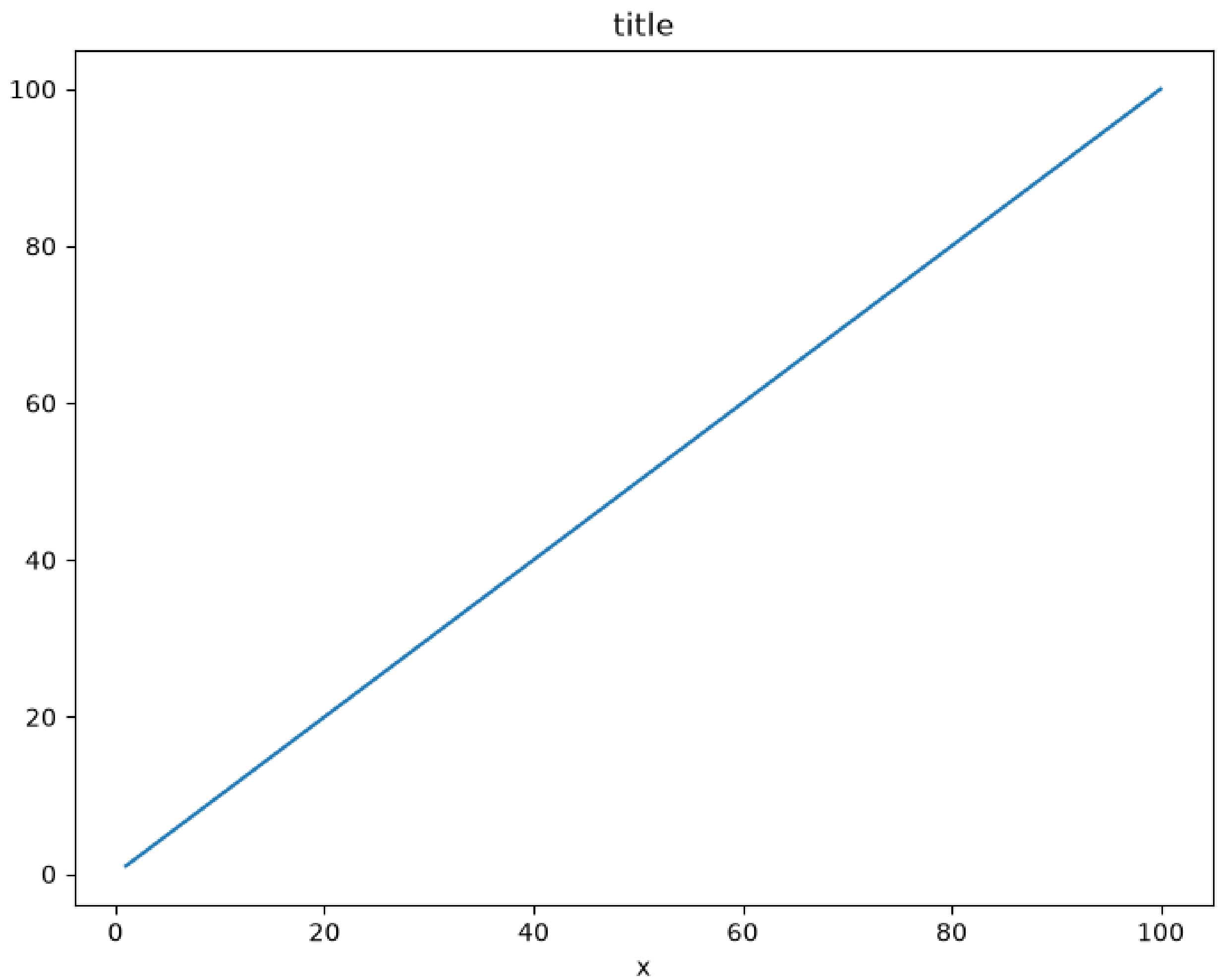
1.Follow along with these steps:

a) Create a figure object called fig using plt.figure()

b) Use add_axes to add an axis to the figure canvas at [0,0,1,1]. Call this new axis ax.

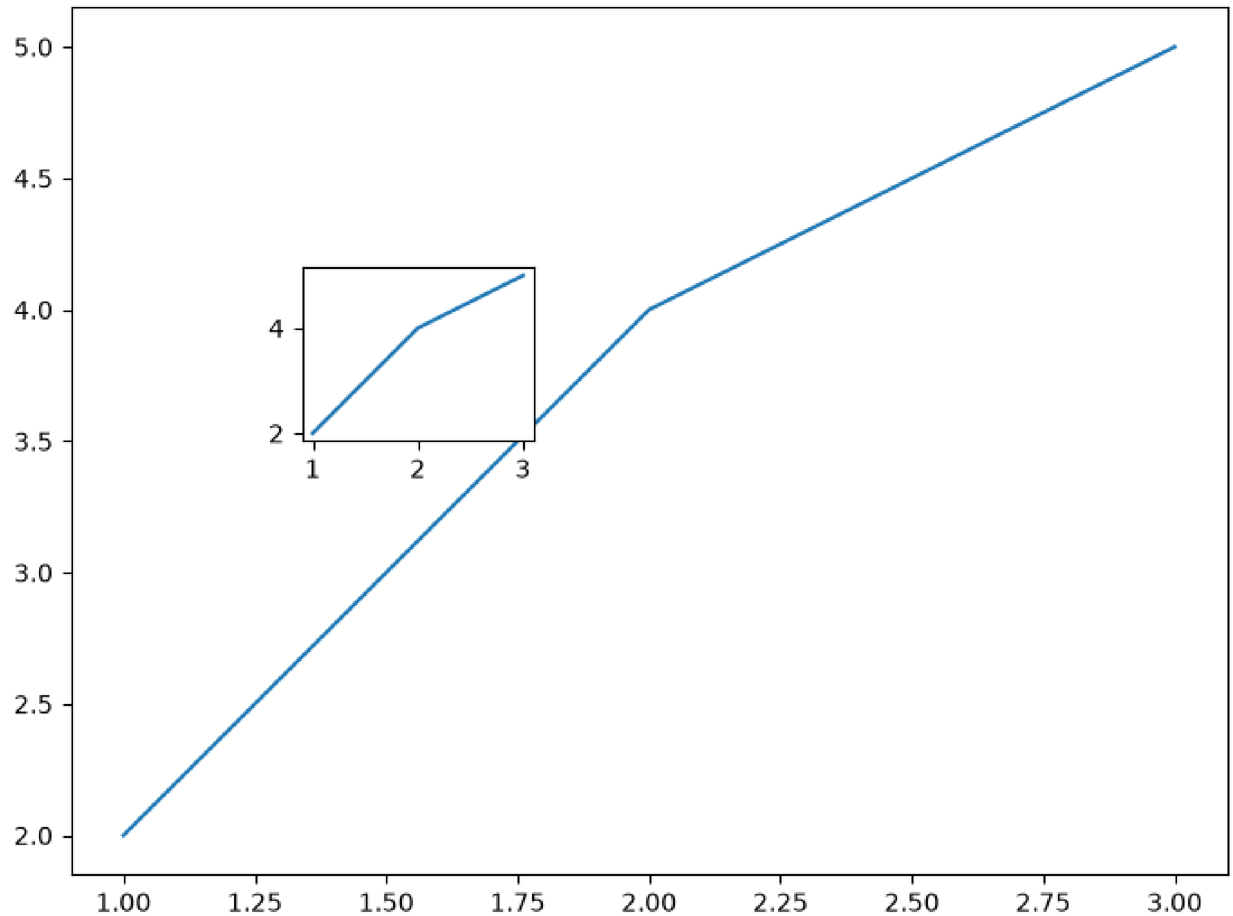
c) Plot (x,y) on that axes and set the labels and titles to match the plot below:

```
fig = plt.figure()
ax =fig.add_axes([0,0,1,1])
ax.set_xlabel("x")
ax.set_title("title")
ax.plot([1,2,100],[1,2,100])
plt.show()
```



Create a figure object and put two axes on it, ax1 and ax2. Located at [0,0,1,1] and [0.2,0.5,.2,.2] respectively. Now plot (x,y) on both axes. And call your figure object to show it.

```
f = plt.figure()
ax1 = f.add_axes([0, 0, 1, 1])
ax2 = f.add_axes([0.2, 0.5, 0.2, 0.2])
x =[1,2,3]
y = [2,4,5]
ax1.plot(x, y)
ax2.plot(x, y)

[<matplotlib.lines.Line2D at 0x7f4f94961cd0>]
```



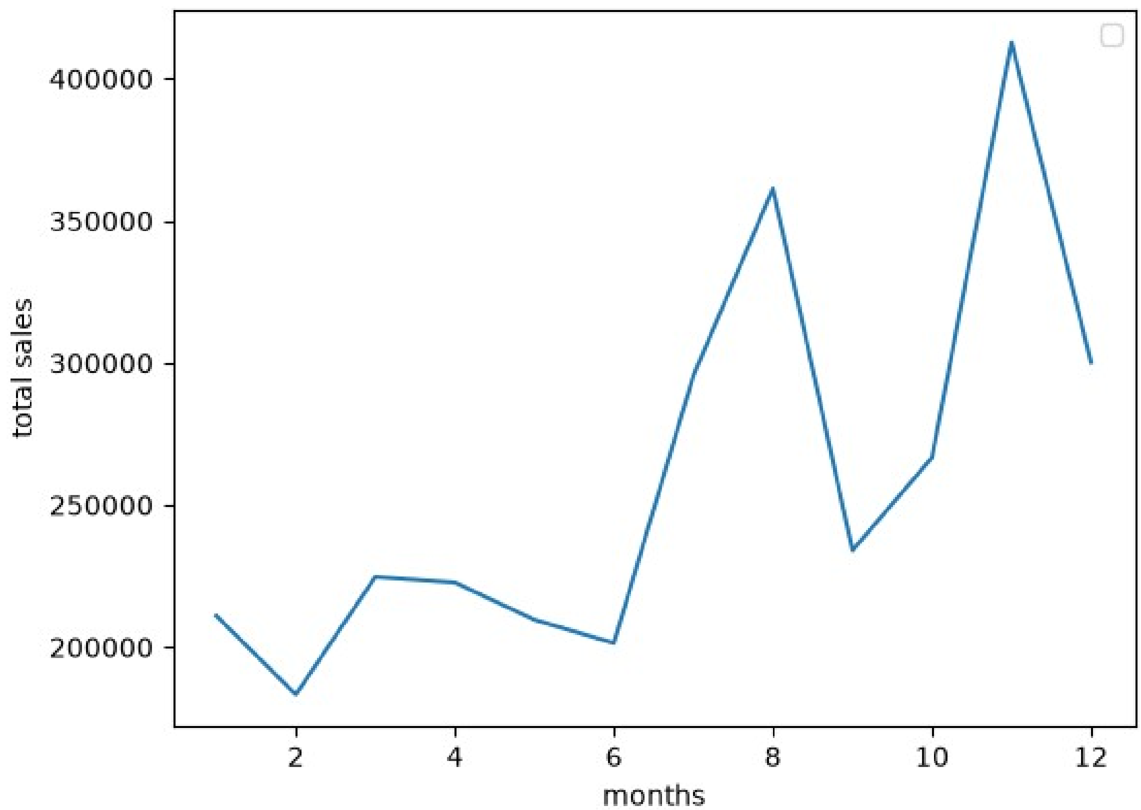
1. Use the company sales dataset csv file, read Total profit of all months and show it using a line plotTotal profit data provided for each month. Generated line plot must include the following properties: –

a. X label name = Month Number

b. Y label name = Total profit

```
data = pd.read_csv("company_sales_data.csv")
plt.plot(data["month_number"],data["total_profit"])
plt.legend()
plt.xlabel("months")
plt.ylabel("total sales")
```

```
/tmp/ipykernel_7516/1077155923.py:3: UserWarning: No artists with
labels found to put in legend.  Note that artists whose label start
with an underscore are ignored when legend() is called with no
argument.
  plt.legend()

Text(0, 0.5, 'total sales')
```



1. Use the company sales dataset csv file, get total profit of all months and show line plot with the following Style properties. Generated line plot must include following Style properties: –

a. Line Style dotted and Line-color should be red

b. Show legend at the lower right location.

c. X label name = Month Number

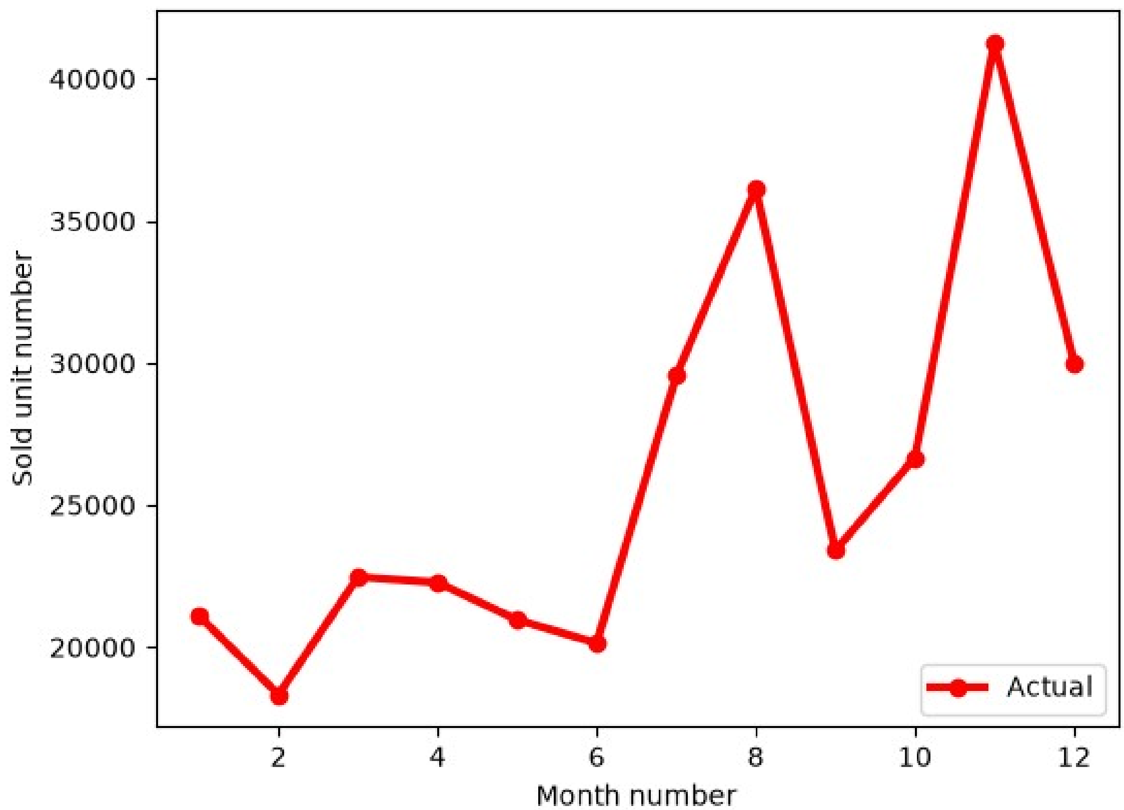
d. Y label name = Sold units number

e. Add a circle marker.

f. Line marker color as read

g. Line width should be 3

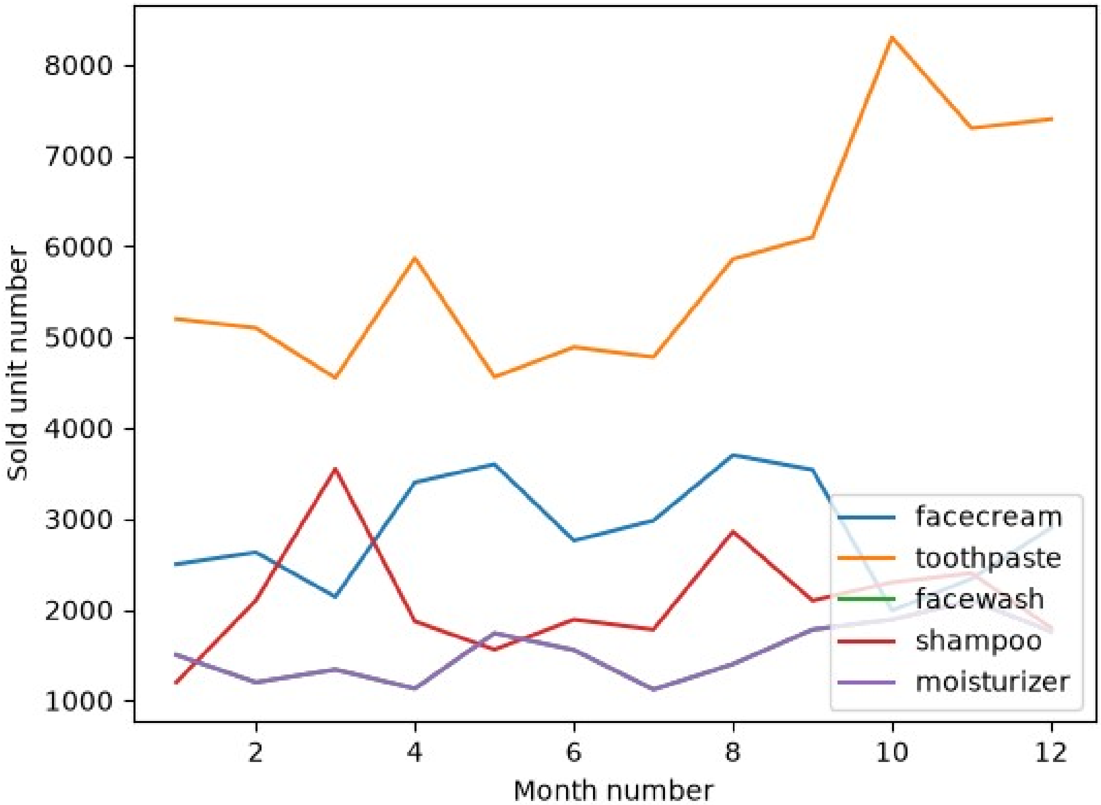
```
data = pd.read_csv("company_sales_data.csv")
plt.plot(data["month_number"],data["total_units"],marker='o',
color='red', linewidth=3, label='Actual')
plt.legend(loc = "lower right")
plt.xlabel("Month number")
plt.ylabel("Sold unit number")

Text(0, 0.5, 'Sold unit number')
```



Additional Questions

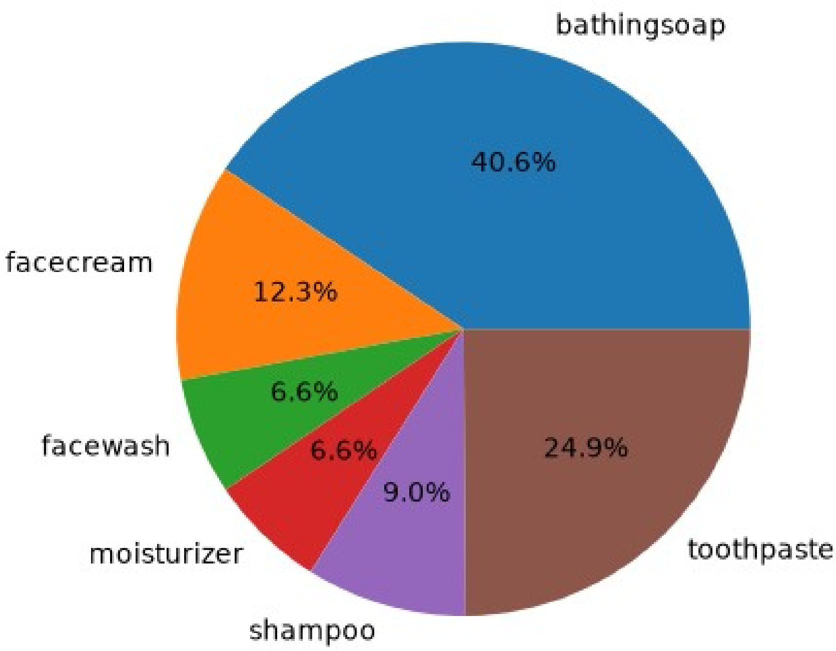
1. Use the company sales dataset csv file, read all product sales data and show it using a multiline plot. Display the number of units sold per month or each product using multiline plots. (i.e., Separate Plotline for each product).

```
plt.plot(data["month_number"],data["facecream"],label='facecream')
plt.plot(data["month_number"],data["toothpaste"],label='toothpaste')
plt.plot(data["month_number"],data["facewash"],label='facewash')
plt.plot(data["month_number"],data["shampoo"],label='shampoo')
plt.plot(data["month_number"],data["moisturizer"],label='moisturizer')
plt.legend(loc = "lower right")
plt.xlabel("Month number")
plt.ylabel("Sold unit number")

Text(0, 0.5, 'Sold unit number')
```



Use the company sales dataset csv file, calculate total sale data for last year for each product and show it using a Pie chart. Note: In Pie chart display Number of units sold per year for each product in percentage.

```
sales =
[data["bathingsoap"].sum(),data["facecream"].sum(),data["facewash"].su
m(),data["moisturizer"].sum(),data["shampoo"].sum(),data["toothpaste"]
.sum()]
labels
=["bathingsoap","facecream","facewash","moisturizer","shampoo","toothp
aste"]
plt.pie(sales,labels=labels,autopct='%1.1f%%')

<matplotlib.container.PieContainer at 0x7f4f9475e4d0>
```


bathingsoap
40.6%
facecream
12.3%
6.6%
facewash
6.6%
9.0%
24.9%
moisturizer
toothpaste
shampoo